osfs26ProjektMS

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Psychology bachelor students' well-being

Introduction

Well-being is an important aspect of mental and physical health. However, there is no commonly agreed definition of this construct [Bayrak, 2024]. Apparently, well-being of graduate students have long been overlooked, unlike the well-being of doctoral students [Schmidt and Hansson, 2018]. Well-being, and health in general, are important prerequisites to successfully meet the demands of studying psychology in a full-time university program. Furthermore, the degree of success at the university can largely determine further career chances, which sometimes start right after the bachelor studies, namely when admission to continue with the master studies is granted only to less than 100 percent of the bachelor students.

One possible indicator of mental and/or physical robust health can be that the bachelor student studies psychology despite having to work to financially support him-/herself. It may also be that these students are better equipped to deal with academic challenges, including stress, compared to students who do not have to work during their bachelor studies. This possibility is what we aimed to investigate.

Methods

Participants and data collection

The sample consisted of a maximum of 12 psychology master students, all of whom attend a current (as of spring 2026) university seminar on the topic 'open science'. The sample consists predominantly of women, which is in line with the generally known sex ratio of psychology students. All participants are relatively young. However, neither sex nor age has been assessed, in order to keep the data as anonymous as possible. An additional way to achieve guaranteed anonymity was that participants entered their answers in an online tabular document at their selected time and place, i.e., not during the seminar time. Last, not least, participation was kindly requested, not coerced.

Preregistration

This study was preregistered (see upload time stamp of this file: click me).

Measures

In total, the main author formulated 15 questions, none of which was based on the research literature on bachelor students' well-being. That is, the questions were formulated only by thinking about what might be related and relevant to (bachelor) students' well-being. However, *a posteriori* superficial comparison with the research literature showed that at least some of the 15 questions were similar to established well-being questionnaire items. For example, financial difficulties and social relationships are often part of these well-being questionnaires [Schmidt and Hansson, 2018]. Of note, this 'study' is of a purely didactic nature, i.e., the only purpose was to collect some relevant data which is of (hopefully) high personal interest to the master students of the 'open science' seminar, in order for them to feel attracted to actively 'play' with what they think might be done with the data in combination with conducting open science.

Research question

Are financial strains during the bachelor study phase positively related to how well academic challenges are dealt with emotionally?

Variables The research question focuses on two of the 15 questions, namely questions 1 and 6, each ranging between values of 1 and 10. Both variables will be treated as if they fulfill the requirements of the interval scale.

Question 1: Financially, it was difficult for me to pay for study fees, rent, and groceries.

Question 6: On average, I felt emotionally capable to deal with the challenges during studying psychology.

Data analysis

A simple linear regression model is used, where the scaled values of questionnaire questions 1 and 6 are the outcome and predictor variable, respectively. The focus will be the 90 percent compatibility interval [Rafi and Greenland, 2020], which will be produced with the R package concurve [Rafi and Vigotsky, 2020]. Additional detailed visualization of the most important part of the Pearson correlation coefficients will be presented using the usual Pearson correlation formula, implemented in the R package correlatio [Miché, 2025].

Note for the students: At this point, I stopped with this manuscript and continued with the dummy R script. After having finished that script and after having uploaded it to GitHub, I will run the code with the original data, and then finish this manuscript. Time now: 14:38 h.

Note 2: Continue with results, time now: 15:26 h.

Results

```r
dat <- data.frame(f1=c(4, 5, 4, 3, 5, 1, 1, 3, 2, 8),
                  f6=c(5, 7, 9, 9, 10, 6, 10, 9, 8, 10))
# Scatterplot and regression line
library(ggplot2)
ggplot(dat) +
    aes(x = scale(f1), y = scale(f6)) +
    geom_point() +
    geom_smooth(color = "blue",
                method = lm, se = FALSE) +
    labs(x="F1", y="F6") +
    theme(
        panel.background = element_blank(),
        axis.text.x=element_text(size=16),
        axis.title.x=element_text(size=16),
        axis.text.y=element_text(size=16),
        axis.title.y = element_text(size=16),
        panel.border = element_rect(color="grey", fill=NA))
```

```r
# Analysis 1: 90 percent compatibility interval
result1 <- cor.test(dat$f1, dat$f6, conf.level = .9, alternative = "greater")
result1$estimate
```

```
##       cor
## 0.2433717
```

```r
result1$conf.int
```

```
## [1] -0.2317387  1.0000000
## attr(,"conf.level")
```

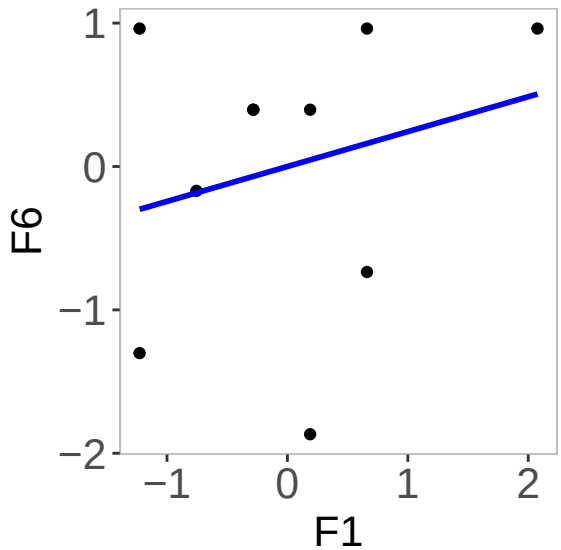

Figure 1: Scatterplot and regression line, both variables being scaled.

```
## [1] 0.9
```

The 90 percent compatibility interval of the directed (right-sided) research question ranges from -.23 to 1.0, with a positive point estimate of the Pearson correlation of .24.

```
# Visualize 90 percent compatibility interval
library(concurve)
# # Help pages of concurve package
# help(package="concurve")
# Use function curve_corr
cor_vis <- curve_corr(x=dat$f1, y=dat$f6, alternative = "greater", method="pearson")
concurve::ggcurve(cor_vis[[1]])
```

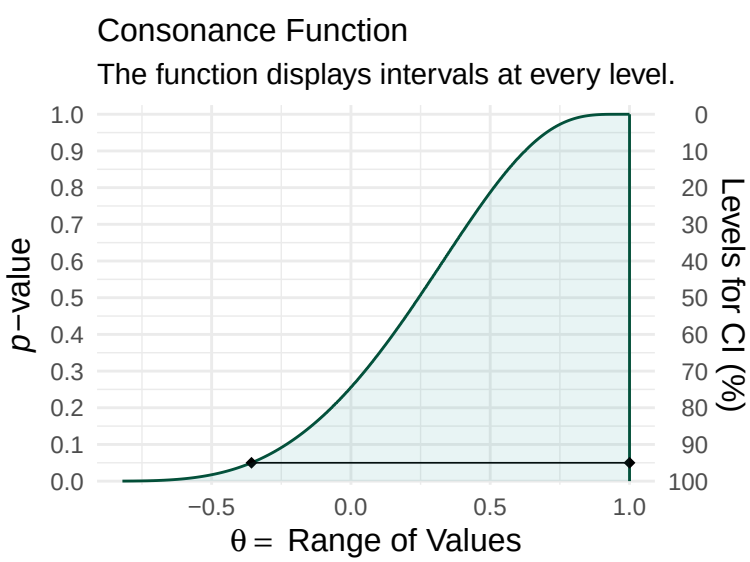

Figure 2: Visualized consonance function of the Pearson correlation result.

The larger part of the 90 percent compatibility interval points into the positive correlation direction (see

Figure 2).

```
library(correlatio)
result2 <- correlatio::corrio(data=dat)
result2$plot2
```

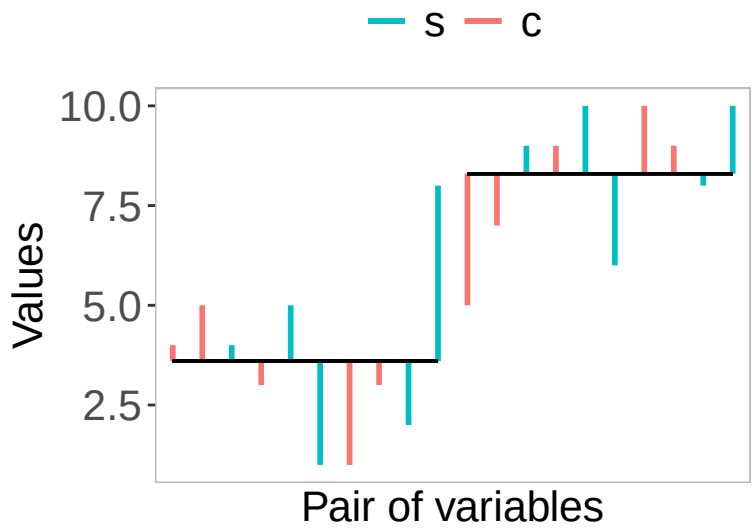

Figure 3: Most important part of Pearson correlation coefficient.

Of the 10 values of both variables, five pairs point into contrary directions (contrary = c = red in Figure 3), whereas the other five pairs point into the same direction (same = s = turquoise in Figure 3). The turquoise pairs have greater pairwise lengths, compared to the pairwise lengths of the red pairs, which is how the positive Pearson correlation coefficient is produced.

Discussion

The expected positive correlation was observed. However, just 10 out of the possible 12 students participated in the questionnaire. The sample size would have been very small, even if all 12 students had participated. Nonetheless, the investigated research question is of interest and may be investigated in the future with a larger sample and with much more diligence regarding how the questions are formulated and what scale to be used.

Limitations

This 'study' has many limitations. First, it is not a serious study, regarding the subject matter. Its actual aim is to trigger the curiosity of the students of wanting to find out, of wanting to use open science, and of wanting to expand their knowledge and their skills, regarding how to actually do all of this; as opposed to passively watching others publishing their findings in fancy ways. Second, the scale of the 15 questions was ill-advised, because the 'middle' did not exist numerically, but it existed verbally. That is, whereas values 1 and 10 contained the extreme labels 'absolutely wrong' and 'absolutely correct', respectively, the value 5 was labelled 'neither nor', which would have been correct for the value 5.5. Third, the sample size is way too low for any empirically serious data analysis. However, this was not of interest in this 'study' (see first limitation).

Strengths

This study has some strengths. First, it refrains from any attempts to immaturely test a statistical hypothesis [Scheel et al., 2021]. Instead, the results are presented and interpreted in a descriptive manner [Shmueli, 2025], which is used only very rarely in academic psychology. Second, the Pearson correlation coefficient is almost always presented only as a single numeric summary of the linear association between two variables.

However, this way, it conceals more than what it reveals, simply because a single numeric summary cannot reveal much. But a visualization of the most important part of the Pearson correlation coefficient can be used to reveal a lot, which, in science, should be the norm, not the exception.

Conclusion

Dear master student of the 'open science' seminar, now it's your turn. Do you accept the challenge?

Note 3: The end, time now 16:23 h.

References

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